

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12397115
Kind Code	B2
Date of Patent	August 26, 2025
Inventor(s)	Wang; De Yu

Injection device

Abstract

An injection device includes a case body, a control circuit disposed on the outside of the case body, a movable part disposed in the case body, carrying a drug, and movable between a start position and a target position, a magnetic member movable along with the movable part and producing a magnetic field, and a magnetic switch disposed on the outside of the case body and connected to the control circuit. When the movable part is moved to the target position, the magnetic switch changes a status of a conductive path of the magnetic switch in response to a presence of the magnetic field. The control circuit changes a state of the injection device according to a change of the status of the conductive path.

Inventors:	Wang; De Yu (Shanghai, CN)
Applicant:	Jabil Circuit (Shanghai) Co., Ltd. (Shanghai, CN)
Family ID:	1000008781780
Assignee:	Jabil Circuit (Shanghai) Co., Ltd. (Shanghai, CN)
Appl. No.:	17/685485
Filed:	March 03, 2022

Prior Publication Data

Document Identifier	Publication Date
US 20220280724 A1	Sep. 08, 2022

Foreign Application Priority Data

CN	202110250319.X	Mar. 08, 2021
----	----------------	---------------

Publication Classification

Int. Cl.: A61M5/20 (20060101); A61M5/178 (20060101); A61M5/24 (20060101); A61M5/31 (20060101); A61M5/315 (20060101)

U.S. Cl.:

CPC A61M5/20 (20130101); A61M5/178 (20130101); A61M5/24 (20130101); A61M5/31 (20130101); A61M5/315 (20130101); A61M5/31546 (20130101); A61M2005/2006 (20130101); A61M2205/14 (20130101); A61M2205/3317 (20130101); A61M2205/3379 (20130101); A61M2205/8212 (20130101)

Field of Classification Search

CPC: A61M (5/20); A61M (5/178); A61M (5/24); A61M (5/31); A61M (5/315); A61M (5/31546); A61M (2005/2006); A61M (2205/14); A61M (2205/3317); A61M (2205/3379); A61M (2205/8212)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
2016/0120751	12/2015	Mounce et al.	N/A	N/A
2018/0353699	12/2017	Helmer	N/A	A61M 5/20
2019/0143051	12/2018	Srinivasan et al.	N/A	N/A
2020/0093993	12/2019	O'Rourke et al.	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
2013148473	12/2012	JP	N/A
2020115031	12/2019	WO	N/A

OTHER PUBLICATIONS

Search Report issued to European counterpart application No. 22160514.0 by the EPO on Jul. 27, 2022. cited by applicant

Primary Examiner: Ahmed; Tasnim Mehjabin

Attorney, Agent or Firm: Young Basile Hanlon & MacFarlane, P.C.

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION

(1) This application claims priority to Chinese Invention Patent Application No. 202110250319.X, filed on Mar. 8, 2021.

TECHNICAL FIELD

(2) The disclosure relates to an injection device, and more particularly to an injection device capable of changing its state when being operated.

BACKGROUND

(3) An injection device is used to administrate a drug or a medicine into a body of a human or an

animal for treatment or wellness promotion. With the development of society and technology, mechanical injection devices no longer satisfy the current needs, and injection devices evolve toward being smart, multifunctional and automated. For example, a detection function is added to an injection device to determine whether a drug to be injected is properly filled in the injection device. However, a conventional injection device with the detection function is still flawed in some ways, such as having a complicated structure and poor reliability.

SUMMARY

(4) Therefore, an object of the disclosure is to provide an injection device that can alleviate at least one of the drawbacks of the prior art.

(5) According to a first aspect of this disclosure, an injection device includes a device case body that has an accommodating space, a control circuit that is disposed on the outside of the device case body, a movable part that is disposed in the accommodating space of the device case body, that carries a drug for injection, and that is movable relative to the device case body between a start position and a target position, a magnetic member that is disposed on the movable part, that is movable along with the movable part, and that produces a magnetic field, and a magnetic switch that is disposed on the outside of the device case body, that is disposed proximate to the magnetic member when the movable part is at the target position, and that is electrically connected to the control circuit. When the movable part is moved to the target position, the magnetic switch changes a status of a conductive path of the magnetic switch in response to presence of the magnetic field that is produced by the magnetic member disposed on the movable part. The control circuit changes a state of the injection device according to a change of the status of the conductive path of the magnetic switch.

(6) According to a second aspect of this disclosure, an injection device includes a device case body that has an accommodating space, a control circuit that is disposed on the outside of the device case body, a movable part that is disposed in the accommodating space of the device case body, and that is movable along a longitudinal direction of the device case body between a start position and a target position, a magnetic member that is disposed on the movable part, that is movable along with the movable part, and that produces a magnetic field, a magnetic switch that is disposed on the outside of the device case body, that is disposed proximate to the magnetic member when the movable part is moved to the target position, and that is electrically connected to the control circuit. The device case body isolates the control circuit and the magnetic switch from the accommodating space of the device case body. When the movable part is moved to the target position, the magnetic switch changes a status of a conductive path of the magnetic switch in response to presence of the magnetic field that is produced by the magnetic member disposed on the movable part. The control circuit changes a state of the injection device according to a change of the status of the conductive path of the magnetic switch.

(7) According to a third aspect of this disclosure, an injection device includes a device case body that has an accommodating space, a control circuit that is disposed on the outside of the device case body, a movable part that is disposed in the accommodating space of the device case body, and that is movable along a longitudinal direction of the device case body between a start position and a target position, a magnetic member that is disposed on the movable part, that is movable along with the movable part, and that produces a magnetic field, a magnetic switch that is disposed on the outside of the device case body, that is disposed proximate to the magnetic member when the movable part is moved to the target position, and that is electrically connected to the control circuit. The device case body isolates the control circuit and the magnetic switch from the accommodating space of the device case body. When the movable part is moved to the target position, the magnetic switch changes a status of a conductive path of the magnetic switch in response to presence of the magnetic field that is produced by the magnetic member disposed on the movable part to pass a signal to the control circuit. The control circuit changes a state of the injection device according to the signal passed by the magnetic switch.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Other features and advantages of the disclosure will become apparent in the following detailed description of the embodiment(s) with reference to the accompanying drawings, of which:
- (2) FIG. 1 is a schematic diagram illustrating an operational concept of an injection device according to one embodiment of the disclosure;
- (3) FIG. 2 is a schematic diagram illustrating a structural concept of the injection device shown in FIG. 1;
- (4) FIG. 3 is a schematic cross-sectional view illustrating an injection device according to one embodiment of the disclosure where a movable part is at a start position;
- (5) FIG. 4 is a schematic cross-sectional view illustrating the injection device shown in FIG. 3 where the movable part is moved to a target position;
- (6) FIG. 5 is a schematic diagram illustrating an operational concept of an injection device according to another embodiment of the disclosure;
- (7) FIG. 6 is a schematic diagram illustrating a structural concept of the injection device shown in FIG. 5;
- (8) FIG. 7 is a schematic diagram illustrating an operational concept of an injection device according to still another embodiment of the disclosure; and
- (9) FIG. 8 is a schematic diagram illustrating a structural concept of the injection device shown in FIG. 7.

DETAILED DESCRIPTION

- (10) Before the disclosure is described in greater detail, it should be noted that where considered appropriate, reference numerals or terminal portions of reference numerals have been repeated among the figures to indicate corresponding or analogous elements, which may optionally have similar characteristics.
- (11) Furthermore, spatially relative terms, such as “on,” “proximate,” “outside,” “inside,” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
- (12) An injection device may adopt a mechanical switch, such as a tactile switch or a miniature snap-action switch, to implement position detection of a movable part, e.g., a plunger, in the injection device, so as to determine whether the movable part has arrived at a target position. In such design, the mechanical switch is disposed inside a case body of the injection device. When the movable part is moved to the target position, the movable part is in physical contact with an actuator nub of the mechanical switch and triggers the mechanical switch to send a signal to a control circuit. In this way, position detection may be realized in the injection device. However, since the mechanical switch is disposed inside the case body and includes a cable which transmits the signal to the control circuit that is disposed on the outside of the case body, the case body needs to be formed with a cable through hole for the cable to pass, increasing complexity of the structure of injection device. Moreover, since the actuator nub of the mechanical switch is disposed in an accommodating space defined by the case body for contact with the movable part which is movable in the accommodating space, the case body needs to be formed with an actuator through hole for the actuator snap to extend into the accommodating space. In this way, liquid drug originally contained in the accommodating space may flow through the actuator through hole into the inside of the case body, and may wet the mechanical switch disposed inside the case body, causing

malfunction or failure of the mechanical switch. In addition, the liquid drug may further flow out of the case body via the cable through hole and wet the control circuit, affecting the function and normal operation of the injection device.

(13) An injection device according to one embodiment of this disclosure includes a device case body having an accommodating space, a control circuit disposed on the outside of the device case body, a movable part disposed in the accommodating space of the device case body, a magnetic member disposed on the movable part, and a magnetic switch disposed on the outside of the device case body. The movable part carries a drug for injection, and is movable relative to the device case body between a start position and a target position. The magnetic member is movable along with the movable part, and produces a magnetic field. The magnetic switch is disposed proximate to the magnetic member when the movable part is moved to the target position, and is electrically connected to the control circuit. When the movable part is moved to the target position, the magnetic switch changes a status of a conductive path of the magnetic switch in response to presence of the magnetic field that is produced by the magnetic member disposed on the movable part. The control circuit changes a state of the injection device according to a change of the status of the conductive path of the magnetic switch.

(14) In some embodiments, the injection device adopts the magnetic switch to detect presence of a magnetic field, so as to determine whether the movable part has arrived at the target position. Since the magnetic switch is not required to be in physical contact with the movable part, the magnetic switch may be disposed on the outside of the device case body, simplifying a structure of the injection device and minimizing the risk that the magnetic switch is affected by liquid, water vapor or other substances in the accommodating space.

(15) Referring to FIGS. 1 and 2, an injection device **10** according to one embodiment of this disclosure includes a device case body **20**, a control circuit **6** that is disposed on the outside of the device case body **20**, a movable part **30** that carries a drug for injection, a magnetic member **40** that is disposed on the movable part **30**, and a magnetic switch **50** that is disposed on the outside of the device case body **20**. The device case body **20** has an accommodating space, and the movable part **30** is disposed in the accommodating space and is movable relative to the device case body **20** between a start position and a target position. The magnetic member **40** is movable along with the movable part **30**, and produces a magnetic field. The magnetic switch **50** is electrically connected to the control circuit **6**, and detects presence of the magnetic field produced by the magnetic member **40** so as to determine whether the movable part **30** has arrived at the target position. In some embodiments, the magnetic member **40** is implemented by a permanent magnet.

(16) Referring to FIG. 2, the device case body **20** is elongated, and includes a closed end and an open end **21** through which a container containing the drug for injection may be put into the injection device **10**. The movable part **30** may be driven by manual operation or a mechanical force to move back and forth in the accommodating space of the device case body **20** between the start position (depicted by solid lines in FIG. 2) and the target position (depicted by broken lines in FIG. 2). In some embodiments, the target position is a ready-to-use position where the drug carried by the movable part **30** is ready to be injected. In other embodiments, the target position may be a specific position where test and inspection are to be performed. The magnetic switch **50** is disposed proximate to the magnetic member **40** when the movable part **30** is moved to the target position.

(17) Referring to FIGS. 3 and 4, a structure of the injection device **10** according to one embodiment of this disclosure is illustrated. The movable part **30** includes a plunger **31** and a cartridge holder **32**. The cartridge holder **32** is used to carry a drug cartridge **33** that contains the drug for injection. In some embodiments, the cartridge holder **32** may be a syringe cartridge, and the drug cartridge **33** may be a prefilled syringe (PFS). In some embodiments, the magnetic member **40** is disposed on the plunger **31**. However, in other embodiments, the magnetic member **40** may be disposed on the cartridge holder **32** or another section of the movable part **30** as long as the magnetic member **40** may be moved together with the movable part **30**. Attributes of the magnetic member **40**, such as,

but not limited to, magnetism, an orientation and a position, are decided based on technical parameters of the magnetic switch **50**. When the movable part **30** is at the target position, the magnetic field applied to the magnetic switch **50** has an intensity higher than a threshold at which the magnetic switch **50** may be switched on, and the magnetic switch **50** changes a status of a conductive path of the magnetic switch **50** in response to the presence of the magnetic field. When the movable part **30** is moved away from the magnetic switch **50**, e.g., retracted to the start position, the magnetic field produced by the magnetic member **40** is not applied to the magnetic switch **50**. In some embodiments, the magnetic member **40** is closer to the magnetic switch **50** when the movable part **30** is at the target position than when the movable part **30** is at the start position.

(18) Referring to FIG. 2, the injection device **10** further includes a circuit case body **60** and a circuit board **70**. The circuit case body **60** is attached to the device case body **20**, is located on the outside of the device case body **20**, and has a receiving space. The circuit board **70** is disposed in the receiving space and holds the control circuit **6**. The magnetic switch **50** is electrically connected to the circuit board **70** and is electrically connected to the control circuit **6** via the circuit board **70**. In some embodiments, the magnetic switch **50** is disposed directly on the circuit board **70**. In other embodiments, the magnetic switch **50** may be separated from the circuit board **70**, and be electrically connected to the circuit board **70** via wired connection or wireless connection. In FIG. 2, the device case body **20** isolates the circuit board **70** and the accommodating space of the device case body **20** from each other, so that the control circuit **6** and the magnetic switch **50** may be isolated from the accommodating space of the device case body **20**. In this way, liquid, water vapor or other substances contained in the accommodating space may be prevented by the device case body **20** from entering the receiving space of the circuit case body **60** and adversely affecting the circuit board **70** and electronic components in the receiving space. In some embodiments, since physical contact between the magnetic member **40** and the magnetic switch **50** is not required, a portion of the device case body **20** where the circuit case body **60** is attached to may have a completely closed structure to prevent substances from entering the receiving space of the circuit case body **60**. In other embodiments, the magnetic switch **50** and/or the control circuit **6** may be embedded within walls of the device case body **20** by a manufacturing process of insert molding, and may be isolated from the accommodating space of the device case body **20**, so as to prevent contact with liquid or other substances in the accommodating space.

(19) In comparison with a mechanical switch, a magnetic switch has a relatively simpler structure, a smaller size, higher operational speed and longer lifespan. In comparison with an electronic switch, a magnetic switch has relatively stronger shock resistance and higher operation reliability. The principle of operation and the structure of a magnetic switch have been well known to a person skilled in the relevant art, and the magnetic switch **50** of the injection device **10** according to this disclosure may be implemented by the MISM-7 series reed switches provided by Littelfuse, Inc. Therefore, detailed descriptions related to the magnetic switch **50** are omitted herein for the sake of brevity. The magnetic switch **50** may be a normally-open reed switch or a normally-closed reed switch. By corresponding circuit design, the two types of reed switches can both realize the functions mentioned in this disclosure.

(20) Referring to FIG. 1, in some embodiments, the control circuit **6** is powered by a battery **75**, and includes an activation circuit **61** and an operation circuit **62**. The activation circuit **61** is a circuit with low power consumption while the operation circuit **62** is a circuit with high power consumption. The magnetic switch **50** is connected to the activation circuit **61**. Before the injection device **10** is put to use, for example, the movable part is at the start position, the injection device **10** is in a standby state where only the activation circuit **61** is in operation and the operation circuit **62** is on standby, so as to achieve a lower power consumption. When the movable part **30** is moved to the target position, the magnetic field produced by the magnetic member **40** causes the magnetic switch **50** to change the status of its conductive path, e.g., changing from disconnection to

connection, or changing from connection to disconnection. The activation circuit **61** changes a state of the injection device **10** according to a change in the status of the conductive path of the magnetic switch **50**. For example, the activation circuit **61** awakes the injection device **10** from the standby state to change the state of the injection device **10**. Specifically, in some embodiments, the magnetic switch **50** changes the status of its conductive path to pass a signal to a controller (not shown) disposed on the circuit board **70**, and the controller controls the activation circuit **61** to change the state of the injection device **10**, i.e., to awake the injection device **10** from the standby state. In some embodiments, awakening the injection device **10** from the standby state includes activating the operation circuit **62** to perform a self-test of the injection device **10** and to supply power to relevant electronic components of the injection device **10**. In some embodiments, the operation circuit **62** may include an indicator, such as a sound indicator or a visual indicator. In an example where the visual indicator is a light-emitting diode (LED) indicator, the LED indicator indicates the state of the injection device **10**. When the operation circuit **62** is activated, the LED indicator is turned on to notify a user that the injection device **10** is awoken from the standby state. (21) It is noted that in some embodiments, the control circuit **6** may be a single circuit which does not include separate activation and operation circuits **61**, **62**. When detecting the change of the status of the conductive path of the magnetic switch **50**, the control circuit **6** starts the self-test, and supplies power to other electronic components of the injection device **10**. In other words, no matter how the control circuit **6** is implemented, the change of the state of the injection device **10** of this disclosure is triggered by the interaction between the magnetic member **40** and the magnetic switch **50**.

(22) Referring to FIGS. **5** and **6**, the injection device **10** according to one embodiment of this disclosure is similar to the injection device **10** shown in FIGS. **3** and **4**, and further includes a magnetic sensor **80**. The magnetic sensor **80** is disposed on the outside of the device case body **20**, is spaced apart from the magnetic switch **50**, and is electrically connected to the control circuit **6**. When the movable part **30** is at the target position, the magnetic field produced by the magnetic member **40** is not only applied to the magnetic switch **50** but also applied to the magnetic sensor **80**. The magnetic sensor **80** measures magnetic field intensity. The control circuit **6** determines, based on the magnetic field intensity measured by the magnetic sensor **80**, whether a magnetic field which causes the magnetic switch **50** to change the status of its conductive path is the magnetic field produced by the magnetic member **40**.

(23) During use, the injection device **10** may be under the influence of an external magnetic field, such as a magnetic field produced by a nearby electronic device, other than the magnetic field produced by the magnetic member **40**. When the external magnetic field exists and when an intensity of the external magnetic field is high enough to change the status of the conductive path of the magnetic switch **50**, the external magnetic field might undesirably awake the injection device **10** from the standby state even if the movable part **30** is not moved to the target position. Therefore, according to the embodiment shown in FIGS. **5** and **6**, the magnetic sensor **80** is introduced for determining a source of a magnetic field, so that a position of the movable part **30** and the state of the injection device **10** may be further verified based on the magnetic field intensity measured by the magnetic sensor **80**.

(24) In some embodiments, when the magnetic field intensity measured by the magnetic sensor **80** is within a predefined range, the control circuit **6** determines that the magnetic field which causes the magnetic switch **50** to change the status of its conductive path is the magnetic field produced by the magnetic member **40**. The predefined range is set in advance based on an intensity of the magnetic field at either pole of the magnetic member **40**. For example, the magnetic sensor **80** may measure an intensity of the magnetic field produced by the magnetic member **40** when the movable part **30** is at the target position under normal operating conditions, and this intensity may serve as a reference value for deciding the predefined range which is a range of plus and minus ten percent of the reference value. On the other hand, when the magnetic field intensity measured by the magnetic

sensor **80** is beyond the predefined range, for example, twice the reference value, the control circuit **6** determines that the magnetic field which causes the magnetic switch **50** to change the status of its conductive path is an external magnetic field rather than the magnetic field produced by the magnetic member **40**. In other embodiments, based on precision requirements of practical application, the predefined range may be flexibly adjusted, for example, to be plus and minus three, five or fifteen percent of the reference value.

(25) It is noted that the magnetic field intensity expected to be measured by the magnetic sensor **80** when the movable part **30** is moved to the target position is not necessarily a fixed value. For example, in one implementation of the injection device, if a relative position of the magnetic sensor **80** to the target position of the movable **30** is somehow changed, the magnetic field intensity expected to be measured may also change. Moreover, if the magnetism of the magnetic member **40** changes, the magnetic field intensity expected to be measured may as well be changed.

(26) Even though the magnetic sensor **80** is configured to measure the magnetic field intensity, the magnetic sensor **80** is not necessarily required to provide an exact value of the magnetic field intensity thus measured to the control circuit **6**. For instance, in some embodiments, the magnetic sensor **80** may be a magnetic sensor chip which acts as a switch; when it is determined by the magnetic sensor chip that the magnetic field intensity thus measured falls in the predefined range, the magnetic sensor chip transmits a notice signal to the control circuit **6** for determination by the control circuit **6** that a source of the magnetic field is the magnetic member **40**. In other embodiments, the magnetic sensor **80** may be a digital magnetic sensor or an analog magnetic sensor which transmits a digital signal or an analog signal that indicates the magnetic field intensity thus measured to the control circuit **6**, to allow the control circuit **6** to determine a source of the magnetic field based on the digital signal or the analog signal.

(27) Referring to FIG. 5, in some embodiments, the magnetic sensor **80** is electrically connected to the operation circuit **62**. The activation circuit **61**, according to the change in the status of the conductive path of the magnetic switch **50**, awakes the injection device **10** from the standby state by activating the operation circuit **62**. Accordingly, the magnetic sensor **80** is powered by the operation circuit **62**, starts measuring an intensity of a magnetic field applied to the magnetic sensor **80**, and provides the intensity thus measured to the operation circuit **62**. When the magnetic field intensity measured by the magnetic sensor **80** is within the predefined range, the operation circuit **62** determines that the magnetic field which causes the magnetic switch **50** to change the status of its conductive path is the magnetic field produced by the magnetic member **40**, and verifies that the movable part **30** has arrived at the target position. When the magnetic field intensity measured by the magnetic sensor **80** is beyond the predefined range, the operation circuit **62** determines that the magnetic field which causes the magnetic switch **50** to change the status of its conductive path is an external magnetic field rather than the magnetic field produced by the magnetic member **40**, and thus shuts down accordingly.

(28) Alternatively, in some embodiments, the magnetic sensor **80** may be directly and electrically connected to the activation circuit **61**. In this arrangement, only when the magnetic switch **50** and the magnetic sensor **80** both meet their respective activation conditions (i.e., the status of the conductive path of the magnetic switch **50** is changed and the magnetic field intensity measured by the magnetic sensor **80** is within the predefined range) will the activation circuit **61** activate the operation circuit **62** to awake the injection device **10**. This arrangement can also realize additional confirmation with respect to the position of the movable part **30** by using the magnetic sensor **80**. In some embodiments as mentioned above where the control circuit **6** is a single circuit and does not include separate activation and operation circuits **61**, **62**, the magnetic sensor **80** may be directly and electrically connected to the control circuit **6**.

(29) Accordingly, it is evident that the magnetic sensor **80** may be utilized to confirm the source of the magnetic field, so that a position of the movable part **30** and the state of the injection device **10** may be further verified based on the magnetic field intensity measured by the magnetic sensor **80**.

A chance that the injection device **10** is undesirably awoken from the standby state by an external magnetic field can be reduced.

(30) In some embodiments, the magnetic sensor **80** includes one of a magnetoresistive sensor, a Hall effect sensor and a combination thereof. Since the magnetoresistive sensor and the Hall effect sensor are existing products available on the market, detailed descriptions therefor are omitted herein for the sake of brevity. For example, the Hall effect sensor of the magnetic sensor **80** may be implemented by the AH180 Hall effect switch provided by Diodes Incorporated. Similar to the magnetic switch **50**, the magnetic sensor **80** is directly disposed on the circuit board **70**, or alternatively, the magnetic sensor **80** may be separated from the circuit board **70** and be electrically connected to the circuit board **70** via wired connection or wireless connection.

(31) Referring to FIGS. **7** and **8**, the injection device **10** according to one embodiment of this disclosure is similar to the injection device **10** shown in FIGS. **5** and **6**, and a difference therebetween resides in that the magnetic member **40** includes a first magnet **41** and a second magnet **42** each of which may be implemented by a permanent magnet. The first magnet **41** produces a first magnetic field. The second magnet **42** is spaced apart from the first magnet **41** and produces a second magnetic field. When the movable part **30** is at the target position, the first magnetic field produced by the first magnet **41** is applied to the magnetic switch **50** without affecting the magnetic sensor **80**, and the second magnetic field produced by the second magnet **42** is applied to the magnetic sensor **80**. When the magnetic field intensity measured by the magnetic sensor **80** is within a preset range, the control circuit **6** determines that a magnetic field which causes the magnetic switch **50** to change the status of its conductive path is the first magnetic field produced by the first magnet **41**. On the other hand, when the magnetic field intensity measured by the magnetic sensor **80** is beyond the preset range, the control circuit **6** determines that the magnetic field which causes the magnetic switch **50** to change the status of its conductive path is an external magnetic field rather than the first magnetic field produced by the first magnet **41**. In some embodiments, the preset range is set in advance based on an intensity of the second magnetic field at either pole of the second magnet **42**. Similar to how the predefined range is set as explained above, the magnetic sensor **80** may measure an intensity of a magnetic field (i.e., the second magnetic field produced by the second magnet **42**) when the movable part **30** is moved to the target position under normal operating conditions, and this intensity thus measured may serve as a reference value for deciding the preset range which is a range of plus and minus a specific percent of the reference value.

(32) In some embodiments, the first magnet **41** is disposed on the movable part **30**, and the first magnet **41** is spaced apart from the magnetic sensor **80** by a predetermined distance when the movable part **30** is at the target position, such that the first magnetic field produced by the first magnet **41** does not affect the magnetic sensor **80**. The predetermined distance may be decided first, with a suitable magnet being then selected based on the predetermined distance to serve as the first magnet **41**. Alternatively, a magnet may be decided first to serve as the first magnet **41**, with the predetermined distance being then found based on the magnetic field distribution of the first magnet **41** to make sure that the first magnetic field would not affect the magnetic sensor **80**.

(33) In some embodiments, referring to FIG. **8**, the injection device **10** further includes a magnetic shielding barrier **90** that is disposed on the outside of the device case body **20**, and that is disposed between the magnetic switch **50** and the magnetic sensor **80** so as to isolate the magnetic sensor **80** from the first magnetic field produced by the first magnet **41** when the movable part **30** is moved to the target position.

(34) By the design that the magnetic member **40** includes the first magnet **41** and the second magnet **42**, and that the magnetic sensor **80** would measure the second magnetic field produced by the second magnet **42** without being affected by the first magnetic field produced by the first magnet **41** when the movable part **30** is at the target position, a higher flexibility in terms of selection and configuration of the magnetic sensor **80** may be achieved. For example, attributes of

the first magnet **41** can be given no consideration when configuring (e.g., selecting) the magnetic sensor **80**, and a location of placement of the magnetic sensor **80** may be adjusted with higher flexibility.

(35) It is noted that in the embodiment as shown in FIGS. **5** and **6**, if the injection device **10** is under the influence of an external magnetic field which is similar to the first magnetic field produced by the first magnet **41**, the external magnetic field would change the status of the conductive path of the magnetic switch **50**, and the intensity of the external magnetic field measured by the magnetic sensor **80** might be within the predefined range. Under such circumstances, the operation circuit **62** would be erroneously activated, and would not determine that the magnetic field resulting in this activation is an external magnetic field so the operation circuit **62** could not shut down successfully. Therefore, the embodiment as shown in FIGS. **7** and **8** is proposed to tackle this issue. The first magnet **41** and the second magnet **42** are configured in such a manner that, when the movable part **30** is at the target position, the magnetic field intensity measured by the magnetic sensor **80** would be that of the second magnetic field produced by the second magnet **42**, and the magnetic field intensity thus measured is lower by a substantial degree (for example, fifty percent or more) than an intensity of the first magnetic field that is capable of causing the magnetic switch **50** to change the status of its conductive path.

(36) With such configurations of the first magnet **41** and the second magnet **42**, the scenario that the operation circuit **62** is erroneously activated and does not shut down successfully because the operation circuit **62** fails to determine that the magnetic field resulting in this activation is an external magnetic field may be avoided. If the injection device **10** shown in FIGS. **7** and **8** is under the influence of an external magnetic field which is similar to the first magnetic field produced by the first magnet **41**, the external magnetic field would change the status of the conductive path of the magnetic switch **50**, and the magnetic field intensity measured by the magnetic sensor **80** would exceed the preset range because the preset range is substantially smaller than the intensity of the first magnetic field and the intensity of such external magnetic field as well. In this way, the control circuit **6** is able to determine that the magnetic field which causes the magnetic switch **50** to change the status of its conductive path is an external magnetic field rather than the first magnetic field produced by the first magnet **41**, so it can be determined that the movable part **30** is not at the target position and the operation circuit **62** can shut down to save power.

(37) It is noted that the magnetic field intensity measured by the magnetic sensor **80** being lower than the intensity of the first magnetic field that is capable of causing a change of a switch status does not necessarily mean that the intensity of the second magnetic field at either pole of the second magnet **42** is lower than the intensity of the first magnetic field at either pole of the first magnet **41**. That is to say, besides magnetism of the first magnet **41** and the second magnet **42**, the preset range may be at least related to relative positions of the magnetic sensor **80** and the second magnet **42**. For example, the preset range is related to distances among the magnetic switch **50**, the magnetic sensor **80**, the first magnet **41** and the second magnet **42**.

(38) To sum up, the injection device **10** according to this disclosure at least has the following advantages. 1. In this disclosure, the magnetic switch **50** is used to replace a mechanical switch so that a structure of the injection device **10** can be improved because through holes for components of a mechanical switch to pass through may be omitted. In addition, the risk that the magnetic switch **50** is affected by liquid, water vapor or other substances may be minimized. 2. In some embodiments, the magnetic sensor **80** is introduced for determination of a source of a magnetic field, so that the position of the movable part **30** and the state of the injection device **10** may be further verified. The chances that the injection device **10** is erroneously awoken may be reduced. 3. In some embodiments, the magnetic member **40** includes the first magnet **41** and the second magnet **42**, and the magnetic sensor **80** measures the second magnetic field produced by the second magnet **42** when the movable part **30** is at the target position, without being affected by the first magnetic field produced by the first magnet **41**. A higher flexibility in terms of selection and

configuration of the magnetic sensor **80** may be achieved. 4. By the specific configurations of the first magnet **41** and the second magnet **42**, such as the intensity of the second magnetic field measured by the magnetic sensor **80** being lower by a substantial degree than the intensity of the first magnetic field that is capable of causing a change of switch state, a scenario that the injection device **10** is erroneously awoken by an external magnetic field similar to the first magnetic field may be further reduced.

(39) In the description above, for the purposes of explanation, numerous specific details have been set forth in order to provide a thorough understanding of the embodiment(s). It will be apparent, however, to one skilled in the art, that one or more other embodiments may be practiced without some of these specific details. It should also be appreciated that reference throughout this specification to “one embodiment,” “an embodiment,” an embodiment with an indication of an ordinal number and so forth means that a particular feature, structure, or characteristic may be included in the practice of the disclosure. It should be further appreciated that in the description, various features are sometimes grouped together in a single embodiment, figure, or description thereof for the purpose of streamlining the disclosure and aiding in the understanding of various inventive aspects, and that one or more features or specific details from one embodiment may be practiced together with one or more features or specific details from another embodiment, where appropriate, in the practice of the disclosure.

(40) While the disclosure has been described in connection with what is (are) considered the exemplary embodiment(s), it is understood that this disclosure is not limited to the disclosed embodiment(s) but is intended to cover various arrangements included within the spirit and scope of the broadest interpretation so as to encompass all such modifications and equivalent arrangements.

Claims

1. An injection device comprising: a device case body having an accommodating space; a control circuit disposed on an outside of the device case body; a movable part disposed in the accommodating space of the device case body, carrying a drug for injection, and movable relative to the device case body between a start position and a target position; a magnetic member disposed on the movable part, movable along with the movable part, and producing a magnetic field; a magnetic switch disposed on the outside of the device case body, disposed proximate to the magnetic member when the movable part is at the target position, and electrically connected to the control circuit; and a magnetic sensor that is disposed on the outside of the device case body, spaced apart from the magnetic switch, and electrically connected to the control circuit, wherein when the movable part is moved to the target position, the magnetic switch changes a status of a conductive path of the magnetic switch in response to presence of the magnetic field that is produced by the magnetic member disposed on the movable part, the control circuit changing a state of the injection device according to a change of the status of the conductive path of the magnetic switch, and wherein when the movable part is moved to the target position, the magnetic field produced by the magnetic member is applied to the magnetic sensor, the magnetic sensor measuring a magnetic field intensity, the control circuit determining, based on the magnetic field intensity measured by the magnetic sensor, whether a magnetic field which causes the magnetic switch to change the status of its conductive path is the magnetic field produced by the magnetic member.

2. The injection device as claimed in claim 1, further comprising: a circuit case body attached to the device case body, located on the outside of the device case body, and having a receiving space; and a circuit board disposed in the receiving space and holding the control circuit; wherein the magnetic switch is electrically connected to the circuit board and is electrically connected to the control circuit via the circuit board.

3. The injection device as claimed in claim 2, wherein the device case body isolates the circuit board and the accommodating space of the device case body from each other.
4. The injection device as claimed in claim 1, wherein the control circuit includes an activation circuit, the magnetic switch is connected to the activation circuit, and when the movable part is moved to the target position, the activation circuit of the control circuit awakes the injection device from a standby state to change the state of the injection device.
5. The injection device as claimed in claim 1, wherein the control circuit includes an activation circuit and an operation circuit, the magnetic switch being electrically connected to the activation circuit, the magnetic sensor being electrically connected to the operation circuit, the activation circuit, according to the change of the status of the conductive path of the magnetic switch, awaking the injection device from a standby state to change the state of the injection device by activating the operation circuit.
6. The injection device as claimed in claim 1, wherein when the magnetic field intensity measured by the magnetic sensor is within a predefined range, the control circuit determines that the magnetic field which causes the magnetic switch to change the status of its conductive path is the magnetic field produced by the magnetic member, the predefined range being set in advance based on an intensity of the magnetic field at either pole of the magnetic member.
7. The injection device as claimed in claim 6, wherein when the magnetic field intensity measured by the magnetic sensor is beyond the predefined range, the control circuit determines that the magnetic field which causes the magnetic switch to change the status of its conductive path is an external magnetic field rather than the magnetic field produced by the magnetic member.
8. The injection device as claimed in claim 1, wherein the magnetic sensor includes one of a magnetoresistive sensor, a Hall effect sensor and a combination thereof.
9. The injection device as claimed in claim 1, wherein the magnetic member includes a first magnet that produces a first magnetic field, and a second magnet that is spaced apart from the first magnet and that produces a second magnetic field, and wherein when the movable part is at the target position, the first magnetic field produced by the first magnet is applied to the magnetic switch without affecting the magnetic sensor, and the second magnetic field produced by the second magnet is applied to the magnetic sensor.
10. The injection device as claimed in claim 9, wherein the first magnet is disposed on the movable part, and the first magnet is spaced apart from the magnetic sensor by a predetermined distance when the movable part is at the target position such that the first magnetic field produced by the first magnet does not affect the magnetic sensor.
11. The injection device as claimed in claim 9, further comprising a magnetic shielding barrier that is disposed on the outside of the device case body, and that is disposed between the magnetic switch and the magnetic sensor so as to isolate the magnetic sensor from the first magnetic field produced by the first magnet when the movable part is at the target position.
12. The injection device as claimed in claim 9, wherein the first magnet and the second magnet are configured in such a manner that, when the movable part is at the target position, the magnetic field intensity measured by the magnetic sensor is that of the second magnetic field produced by the second magnet and is lower than an intensity of the first magnetic field that is capable of causing the magnetic switch to change the status of its conductive path.
13. The injection device as claimed in claim 12, wherein the intensity measured by the magnetic sensor is at least fifty percent lower than an intensity of the first magnetic field that is capable of causing the magnetic switch to change the status of its conductive path when the movable part is at the target position.
14. The injection device as claimed in claim 12, wherein when the magnetic field intensity measured by the magnetic sensor is within a preset range, the control circuit determines that a magnetic field which causes the magnetic switch to change the status of its conductive path is the first magnetic field produced by the first magnet, when the magnetic field intensity measured by

the magnetic sensor is beyond the preset range, the control circuit determines that the magnetic field which causes the magnetic switch to change the status of its conductive path is an external magnetic field rather than the first magnetic field produced by the first magnet, and the preset range is set in advance based on an intensity of the second magnetic field at either pole of the second magnet.

15. The injection device as claimed in claim 1, wherein the movable part includes a plunger, and the magnetic member is disposed on the plunger.

16. The injection device as claimed in claim 1, wherein the target position is a ready-to-use position where the drug carried by the movable part is ready to be injected.

17. An injection device comprising: a device case body having an accommodating space; a control circuit disposed on an outside of the device case body; a movable part disposed in the accommodating space of the device case body, and movable along a longitudinal direction of the device case body between a start position and a target position; a magnetic member disposed on the movable part, movable along with the movable part, and producing a magnetic field; a magnetic switch disposed on the outside of the device case body, disposed proximate to the magnetic member when the movable part is moved to the target position, and electrically connected to the control circuit; and a magnetic sensor disposed on the outside of the device case body, spaced apart from the magnetic switch, and electrically connected to the control circuit, wherein the device case body isolates the control circuit and the magnetic switch from the accommodating space of the device case body; wherein when the movable part is moved to the target position, the magnetic switch changes a status of a conductive path of the magnetic switch in response to presence of the magnetic field that is produced by the magnetic member disposed on the movable part, the control circuit changing a state of the injection device according to a change of the status of the conductive path of the magnetic switch, and wherein when the movable part is moved to the target position, the magnetic field produced by the magnetic member is applied to the magnetic sensor, the magnetic sensor measuring a magnetic field intensity, the control circuit determining, based on the magnetic field intensity measured by the magnetic sensor, whether a magnetic field which causes the magnetic switch to change the status of its conductive path is the magnetic field produced by the magnetic member.

18. An injection device comprising: a device case body having an accommodating space; a control circuit disposed on an outside of the device case body; a movable part disposed in the accommodating space of the device case body, and movable along a longitudinal direction of the device case body between a start position and a target position; a magnetic member disposed on the movable part, movable along with the movable part, and producing a magnetic field; a magnetic switch disposed on the outside of the device case body, disposed proximate to the magnetic member when the movable part is moved to the target position, and electrically connected to the control circuit; and a magnetic sensor disposed on the outside of the device case body, spaced apart from the magnetic switch, and electrically connected to the control circuit, wherein the device case body isolates the control circuit and the magnetic switch from the accommodating space of the device case body; wherein when the movable part is moved to the target position, the magnetic switch changes a status of a conductive path of the magnetic switch in response to presence of the magnetic field that is produced by the magnetic member disposed on the movable part to pass a signal to the control circuit, the control circuit changing a state of the injection device according to the signal passed by the magnetic switch; and wherein when the movable part is moved to the target position, the magnetic field produced by the magnetic member is applied to the magnetic sensor, the magnetic sensor measuring a magnetic field intensity, the control circuit determining, based on the magnetic field intensity measured by the magnetic sensor, whether a magnetic field which causes the magnetic switch to change the status of its conductive path is the magnetic field produced by the magnetic member.
